

	$v = -142.86 \approx 143 \text{ m s}^{-1}$												
1(c)(ii)	Mass of gas after 15 s = $4000 - (15.0 \times 70.0) = 2950 \text{ kg}$ Resultant force = ma $T - mg = ma$ $10000 - 2950(1.62) = 2950a$ $a = 1.7698 \approx 1.77 \text{ m s}^{-2}$												
1(c)(iii)	As rocket has been fired for 15 s, its total mass decreases since fuel is used. The resultant force being (thrust – weight) will thus increase. Hence the acceleration increases.												
2(a)	$R = \frac{\rho L}{A}$ $\rho = \left(\frac{R}{L}\right)A$ $= \left(\frac{R}{L}\right)(\pi r^2)$ $= (1.73)\pi \left(\frac{1.02 \times 10^{-3}}{2}\right)^2$ $= 1.4136 \times 10^{-6}$ $= 1.41 \times 10^{-6} \Omega\text{m}$												
2(b)(i)	<table border="1"><thead><tr><th>time after switched on/s</th><th>ΔU</th><th>q</th><th>w</th></tr></thead><tbody><tr><td>0 – 59</td><td>positive</td><td>negative</td><td>positive</td></tr><tr><td>60 – 100</td><td>zero</td><td>negative</td><td>positive</td></tr></tbody></table> <u>For first 60s,</u> Work done on the wire by the power supply is positive. Temperature increases, molecular KE and hence internal energy increase. Since temperature of the wire is higher than room temperature, heat supplied to the system is negative. <u>From 60 s to 100s,</u> Work done on the wire by the power supply is positive. Temperature remain constant, molecular KE and hence internal energy remain unchanged. Since temperature of the wire is higher than room temperature, heat supplied to the system is negative.	time after switched on/s	ΔU	q	w	0 – 59	positive	negative	positive	60 – 100	zero	negative	positive
time after switched on/s	ΔU	q	w										
0 – 59	positive	negative	positive										
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No.	Solution
2(b)(ii)	